

Biochar-Enhanced Microbial Fuel Cells (MFCs) with Battery Management System

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1 Abstract

Microbial Fuel Cells (MFCs) are bioelectrochemical systems that convert microbial metabolic activity into electrical energy. This work explores the use of **biochar-based electrodes** to enhance electron transfer and improve system sustainability. Biochar, derived from organic waste via pyrolysis, provides high surface area and conductivity, enabling improved microbial adhesion and performance.

Due to inherently low and unstable MFC output, a **Battery Management System (BMS)** is integrated for energy storage and regulation. The combined system demonstrates a practical waste-to-energy solution by powering a low-power RC device, highlighting its potential for decentralized renewable energy applications.

2 Introduction

Growing energy demand and wastewater pollution necessitate sustainable technologies that can address both challenges simultaneously. MFCs offer a promising solution by generating electricity while treating wastewater.

However, traditional electrode materials such as platinum and graphite are expensive and limit scalability. Additionally, MFC output is often insufficient for direct application. This study addresses these limitations by integrating biochar electrodes and a power management system.

3 Problem Statement

- High cost of conventional electrode materials limits scalability.
- Low and unstable electrical output of MFCs restricts practical usage.
- Lack of integrated systems for energy storage and utilization.

4 Literature Review

- Conventional MFCs use platinum/graphite electrodes with high efficiency but poor scalability.
- Biochar has shown promise in energy systems due to high surface area and conductivity.
- Recent studies indicate improved performance with biochar electrodes, though optimization challenges remain.

5 Methodology

5.1 Biochar Preparation

Organic waste is pyrolyzed at controlled temperatures to produce biochar with desired porosity and conductivity.

5.2 MFC Construction

A dual-chamber MFC is fabricated using biochar electrodes. Microbial cultures are introduced to facilitate organic matter degradation.

5.3 System Integration

A Battery Management System (BMS) is integrated to store and regulate electrical output.

5.4 Performance Evaluation

System performance is evaluated based on voltage output, power density, and stability.

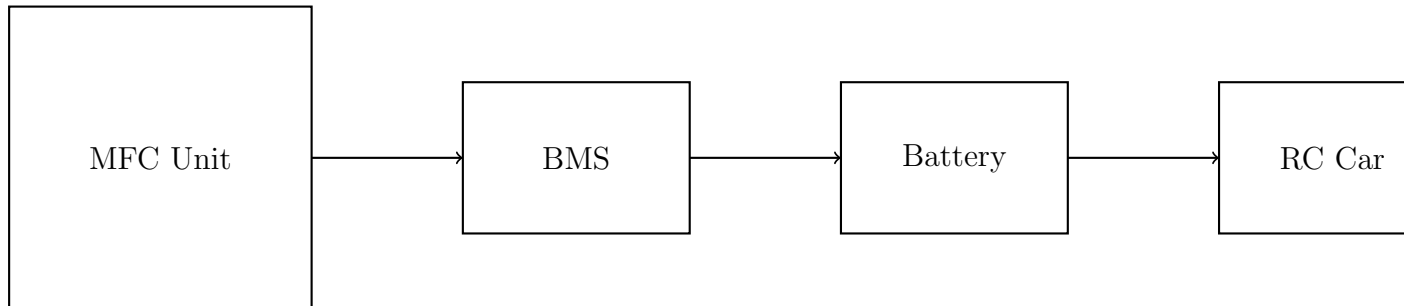
6 Working Principle

$$P = VI$$

$$\eta = \frac{P_{output}}{P_{input}} \times 100$$

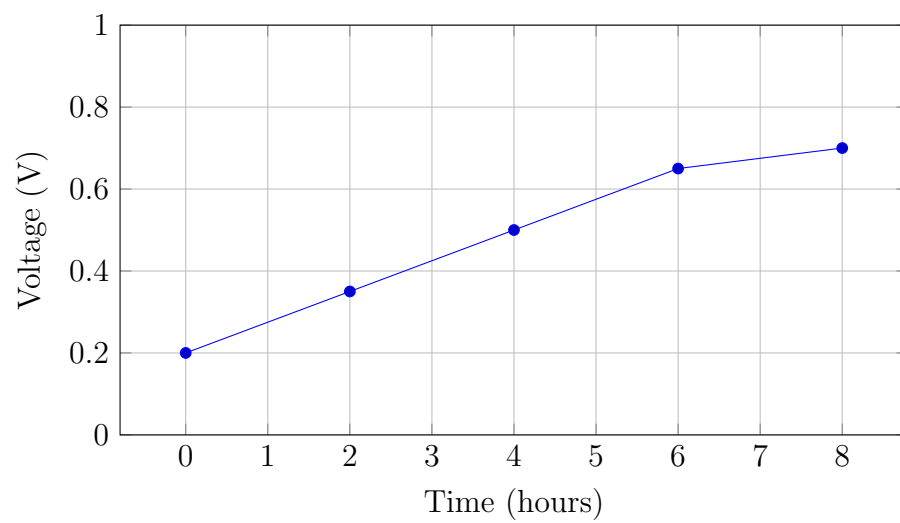
Electrons generated from microbial metabolism travel through the external circuit, generating current, while protons migrate through the membrane to complete the circuit.

7 System Architecture

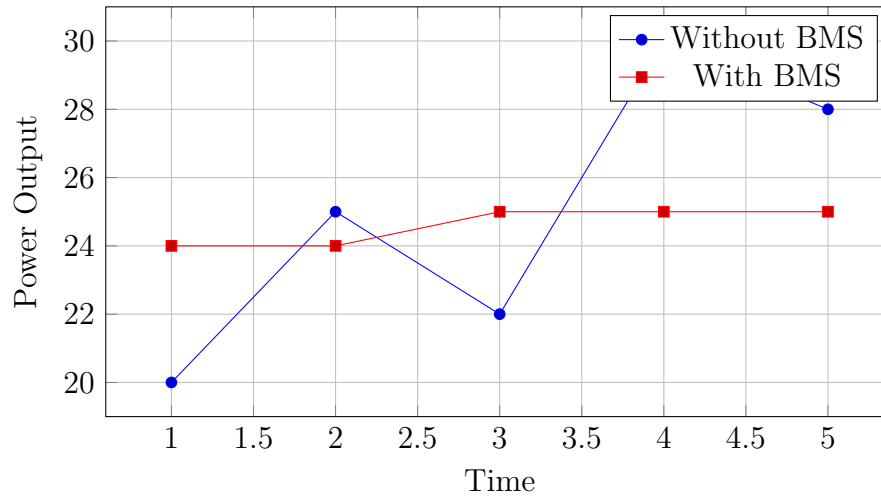


8 Results and Discussion

8.1 Voltage Output



8.2 Power Stability with BMS



9 Advantages

- Sustainable and eco-friendly
- Cost-effective electrode material
- Improved microbial activity and electron transfer
- Integrated energy storage system
- Waste-to-energy capability

10 Limitations

- Low power density
- Electrode degradation over time
- Performance variability based on conditions

11 Scalability and Feasibility

Biochar production from agricultural waste ensures scalability. Multiple MFC units can be connected in series/parallel to increase output. The system is suitable for decentralized wastewater treatment and low-power applications.

12 Conclusion

This study demonstrates a sustainable approach to energy generation using biochar-enhanced MFCs integrated with a BMS. The system offers a practical, scalable, and eco-friendly solution for simultaneous wastewater treatment and electricity generation.